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Gender differences in grit and programming experience across course levels and their relationship to undergraduate students' intention to complete a computer science degree

Oluwafemi Osho[a], Ananya Gupta[a], Devon Brown[b], Bart Kninjenburg[a], Cazembe Kennedy[c], Eileen Kraemer[a], Stacey Sexton[d], Gloria Washington[b], Prakriti Subedi[b], Nathan Demssie[e], Essence Kelsey[f], John Porter III[g], Kinnis Gosha[g] and Simone A. Smarr[b]

[a]School of Computing, Clemson University, Clemson, SC, USA; [b]Department of Electrical Engineering and Computer Science, Howard University, Washington, DC, USA; [c]Peabody Library and College of Connected Computing, Vanderbilt University, Nashville, TN, USA; [d]Independent Consultant; [e]Computer Science Department, Columbia University, New York, NY, USA; [f]Computer Science Department, Spelman College, Atlanta, GA, USA; [g]Computer Science Department, Morehouse College, Atlanta, GA, USA

ABSTRACT

Background and Context: Despite the growing demand for computing professionals, high dropout rates and pronounced gender disparities persist in CS programs. While factors such as grit and programming experience influence student retention and success and may differ by gender, how these differences shape retention remains unclear.

Objective: This paper employs the Social Cognitive Career Theory (SCCT) to study how gender differences in grit and programming experience across the introductory computing curriculum affect students' intention to complete their computer science (CS) degree.

Method: We collected survey data from 303 undergraduate CS major students enrolled in CS1, CS2, and CS3 at an engineering-focused university in the southeastern United States.

Findings: Results showed differences between men and women at different course levels in terms of grit and programming experience, which in turn influenced their self-efficacy to study CS, outcome expectations, attitude toward computing, and consequently their intention to continue studying CS as their major.

Implications: The study highlights the need for intervention strategies to enhance retention and provide tailored, gender-inclusive support for computer science students.

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1. Introduction

The computing industry has been one of the fastest-growing sectors in recent years, leading to a growing demand for computing professionals (Giannakos et al., 2017). Despite increased demand, attrition of students from computer science (CS) programs occurs due to a high dropout rate from the program, with the majority of

CONTACT Oluwafemi Osho oosho@clemson.edu School of Computing, Clemson University, McAdams Hall, 821 McMillan Rd, Clemson, SC 29634, USA

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student withdrawals taking place within the first two years of study (Giannakos et al., 2017; Osho et al., 2022). The high dropout predicament is compounded by a significant gender disparity among students choosing to pursue or complete computer science degrees at the postsecondary level, with the balance tilted heavily in favor of men in terms of degree completion (Main & Schimpf, 2017; Nolan et al., 2019; Rubio et al., 2015). This pattern contrasts with broader trends in higher education, where women surpass men in overall postsecondary degree completion (Katrňák et al., 2025; Wells et al., 2011). In fact, among all fields of science, technology, engineering, and mathematics (STEM), computer science is the only field in which women's representation has steadily declined over the past few decades (Cohoon & Aspray, 2006; Corbett & Hill, 2015; Wu & Uttal, 2024). In the United States, for example, the proportion of women earning CS bachelor's degrees dropped from 35% in the mid-1980s (Cohoon, 2011; Wu & Uttal, 2024) to 21.9% in 2021 (Wu & Uttal, 2024). Similar downward trends have been documented in Canada, New Zealand, the European Union, and Australia (Klawe, 2011; Vitores & Gil-Juárez, 2016). Furthermore, women are approximately twice as likely as men to consider leaving a computer science major (Kapoor & Gardner McCune, 2018). In line with this evidence, one study found that women were less likely than men to remain in computing majors two years after completing the introductory CS course (Lehman et al., 2023). The continued underrepresentation of women in computer science reinforces gender disparities in the technology workforce, which undermines productivity, stifles innovation, and reduces competitiveness (Frachtenberg & Kaner, 2022; Kordaki & Berdousis, 2020). Having more women in the computing field will promote equity and diversity, ensure a competitive workforce, and empower women to participate in contributing to technical innovations (Ashcraft et al., 2012; Cohoon, 2002; Frachtenberg & Kaner, 2022; Kordaki & Berdousis, 2020; Main & Schimpf, 2017).

Generally, when students drop out of computing programs, it potentially leads to a shortage of qualified individuals needed to meet the technology industry's demands (Schefer-Wenzl et al., 2023). Consequently, researchers have attempted to understand the factors connected to student retention and success in computing courses. Prior work emphasizes the significance of factors such as grit – steady passion and persistent effort toward long-term goals (Duckworth et al., 2007) – and programming experience directly or indirectly influencing the retention and graduation of computing majors (Lishinski et al., 2022; Osho et al., 2024; Roberts et al., 2011). Regrettably, notable gender differences are found in programming experience, with women generally reporting significantly less programming experience than men (Beyer et al., 2005; Murphy et al., 2006; Osho et al., 2022; Rankin & Thomas, 2020). The deficit in programming experience among women is more likely attributable to systemic factors than personal inadequacies. An analysis of studies on student recruitment and learning in computing majors revealed that women in undergraduate CS programs enroll in their major because it aligns with their strengths and abilities (Cohoon, 2002). They also perceive the difficulty of programming concepts similarly to men and catch up in mastering programming concepts by the time they complete their introductory CS classes (Murphy et al., 2006). On average, however, women who pursue computing degrees do not begin programming until college, while men start in high school (Beyer et al., 2005; Murphy et al., 2006). This delayed exposure can contribute to feelings of inadequacy.

Moreover, women often face other systemic challenges in computer science programs, including pervasive stereotypes and a lack of continuous encouragement and support from family, friends, and faculty, compared to men (Cheryan et al., 2009, 2015, 2017; Cohoon, 2002; Hunt et al., 2022; Watson, 2020). Due to these stereotypes, women students more strongly perceive that their abilities are underestimated by men faculty members in computer science (Murphy et al., 2006), which can further erode their confidence and persistence in the field. Collectively, these systemic issues suggest that women might require higher levels of resilience (a component of grit) to persist and succeed in the field. Although some studies demonstrate gender differences in grit and/or its sub-components (e.g. Christensen & Knezek, 2014; Kaya & Yüksel, 2022; Sigmundsson & Leversen, 2024), it remains unclear how these differences influence retention. A comprehensive understanding of how grit and programming experience interact across different genders and academic levels is essential for devising effective strategies to enhance retention and provide tailored, gender-appropriate support for computer science students.

Our study examines student retention in CS programs by concentrating on the factors that shape students' intention to complete their degrees. To guide our inquiry, we employ Social Cognitive Career Theory (SCCT) (Lent et al., 1994, 2002), a widely used framework for understanding the development and execution of career- and academic-related interests and decisions (Liguori et al., 2020; Wofford, 2021). The SCCT framework postulates that personal attributes (e.g. gender) and contextual factors (e.g. experience) interact with motivational mechanisms (e.g. self-efficacy beliefs and outcome expectations) to influence individuals in pursuing their goals or intentions (Lent et al., 1994, 2001, 2002; Wang et al., 2022). The theory has been applied to learning experiences and vocational interests among underrepresented student groups (Dickinson et al., 2017), engineering students' academic goals (Lent, Sheu, et al., 2008), and gender differences in students' intention to major in computer science (Sax et al., 2017). In the present study, first, we elaborate on the SCCT framework by including grit and programming experience to study how gender differences in these constructs, together with attitude toward CS, relate to undergraduate CS students' intention to complete their degree. Second, given that women often start their degree with a deficit but catch up over the course of the introductory CS classes (Murphy et al., 2006), we investigate how these relationships play out across the introductory computing curriculum. In specific terms, this study answers the following questions:

(i) Are there gender differences in students' grit and programming experience?
(ii) How do gender differences in grit and current programming experience between men and women relate to their intention to complete their CS degree?
(iii) Do these differences between men and women vary by course level?

1.1. Contributions

Our work demonstrates how retention factors in CS education – specifically, grit and programming experience – differ by gender. It adds empirical evidence to our understanding of how these factors interact and impact students' educational experiences and outcomes. Second, we discuss how our findings can inform interventions, policies, and

practices that aim to address gender disparities and support the success of women in computer science.

The remainder of this paper is structured as follows: In Section 2, we outline a theoretical framework integrating related work on grit, computer programming experience, and gender differences in CS education into the social cognitive career theory (SCCT) framework. This theoretical framework motivates our research model described in Section 3. After describing our positionality in Section 4 and our survey research methods in Section 5, Section 6 presents a path analysis of our research model using our survey data. The consequences, implications, and limitations of these findings are presented in Section 7, which is followed by our conclusion in Section 8.

2. Background and theoretical framework

In this section, we review the theoretical and empirical foundations that guide our research, beginning with Social Cognitive Career Theory (SCCT) (Lent et al., 1994, 2002) as our primary framework. We then explore the literature on gender disparities in CS education, grit, and the influence of programming experience within the broader context of computer science.

2.1. Social cognitive career theory

Social Cognitive Career Theory (SCCT) (Lent et al., 1994, 2002) provides an integrative framework that explains the processes and mechanisms involved in the development of career and academic interests, the formation and implementation of career-related decisions, and the attainment of performance outcomes. Rooted in Bandura's Social Cognitive Theory (Bandura, 1986, 1997), SCCT is widely recognized and validated as a leading model for understanding the psychological processes that drive career interests and goals (Liguori et al., 2020). Given the term "career" within the Social Cognitive Career Theory (SCCT) and the preponderance of career-related studies applying this framework (e.g. Aure et al., 2019; Austin & Nauta, 2016; Belchior & Lyons, 2021; Brändle & Kuckertz, 2022; Liguori et al., 2020; Pfeifer et al., 2016; Pham et al., 2024; Tran & Von Korflesch, 2016; Uansa-Ard & Wannamakok, 2022; Udayanan, 2019), it may be tempting to conclude that SCCT solely applies to career-related research. However, Lent and his co-authors, the developers of SCCT, conceptualized the theory to address both career and academic behaviors (Lent et al., 1994, 2002). This broader scope allows SCCT to be utilized effectively to understand various aspects of individual development and decision-making in both career and educational contexts, including academic interests, choices, and performance.

At the core of the SCCT framework are three social cognitive mechanisms pertinent to academic and career development: self-efficacy beliefs, outcome expectations, and goal representations. Self-efficacy is defined as "people's judgments of their capacities to organize and execute courses of action required to attain designated types of performances" (Bandura, 1986, p. 391). Self-efficacy beliefs are a dynamic trait concerned with a person's judgment of their capabilities (e.g. "Can I do this?"), which may influence their choice of activities, effort expenditure, thought patterns, and reactions to obstacles (Lent

et al., 1994). In addition, self-efficacy has been found to predict academic and career-related goals and intentions (Lent et al., 1994; Liguori et al., 2020).

Whereas self-efficacy is concerned with individuals' perceived capabilities, outcome expectations refer to the anticipated consequences of engaging in specific behaviors (e.g. "What will happen if I do this?") (Lent et al., 1994). According to Bandura (1986), the anticipated outcomes can be physical (e.g. high-paying job opportunities in the technology industry), social (e.g. praise and support from peers and family), or self-evaluative (e.g. self-satisfaction). Outcome expectations strongly influence goals or intentions (Liguori et al., 2020). Both self-efficacy and outcome expectations are acquired and modified through learning experiences (Gregory & Penela, 2023; Lent et al., 1994, 2002).

Goal representations encode a person's desire to undertake a specific activity, engage in a particular behavior, or achieve a certain outcome (Lent et al., 1994, 2002, 2017; Liguori et al., 2020; Wang et al., 2022). According to the social cognitive theory, goals are essential in self-regulating behavior and maintaining a reciprocal relationship with self-efficacy and outcome expectations (Bandura, 1986; Lent et al., 1994): On the one hand, goals provide direction and motivation for individuals' actions, enhance their self-efficacy, and shape their outcome expectations. On the other hand, goals are influenced by individuals' self-efficacy beliefs and outcome expectations.

The SCCT framework posits that personal characteristics (e.g. gender) and contextual variables (e.g. experience) interrelate with self-efficacy beliefs and outcome expectations to pursue goals or intentions (Lent et al., 1994, 2001, 2002; Wang et al., 2022). Personal characteristics encapsulate an individual's personality and predispositions; contextual factors connote past or recent experiences that influence an individual's perceptions, beliefs, and overall experiences (Liguori et al., 2020).

Within the computing discipline, scholars have applied SCCT to study factors that motivate students to pursue a CS college degree (Alshahrani et al., 2018), impact high school students' decisions to select an IT-related major (Luse et al., 2014), and influence Spanish high school students' interest in technology (Inda-Caro et al., 2016). However, research on students' behavioral intention to persist in their computing programs remains limited. Our review identified only a few studies on this topic (e.g. Lehman et al., 2023; Lent et al., 2011; Lent, Lopez, et al., 2008). The latter two applied the SCCT choice model to investigate how contextual support and barriers act as sources of social cognitive mechanisms that influence students' interest and goal of persisting in the computing discipline.

The current study advances prior research on Social Cognitive Career Theory (SCCT) by applying it to the field of Computer Science, a narrower computing field where women and minority groups are still significantly underrepresented. Moreover, it introduces variables such as grit and programming experience, which have not been previously examined within the SCCT framework, adding new dimensions to understanding factors that influence students' retention and success in the CS domain.

2.2. Gender differences in CS education

While several factors play a role in student success and retention in computing, it is essential to recognize that individuals from diverse backgrounds and experiences

may experience these factors differently. For example, a meta-analysis of 187 studies on gender differences in academic self-efficacy by Huang (2013) revealed that women exhibited lower computer self-efficacy than men. A study by Osho et al. (2024) showed that while women had higher levels of belief in the struggle of computer scientists, they reported lower levels of programming experience, attitude toward CS, intention to complete their computing program, computing identity, and self-efficacy.

These results warrant a further unpacking of the root causes of these gender differences. Notably, the early deficit-based narrative around women in computing has been thoroughly debunked. For instance, according to Gibb et al. (2008), women consistently outperform men on standardized tests and attain more school and post-school qualifications. In their study, teacher assessments indicated that men exhibited higher levels of inattentiveness, restlessness, and distractibility, as well as a tendency toward aggressive, antisocial, and oppositional behaviors compared to women. These findings can be explained by a study conducted by Warrington et al. (2000), which found that while societal expectations differ for boys and girls regarding academic achievement, girls were more accepted for working hard and still being part of the "in crowd", whereas boys faced pressure to conform to a "cool", masculine image and were more likely to be ridiculed for working hard.

The underrepresentation of women in CS must thus be sought in discrepancies in subjective experiences rather than a lack of ability. For instance, based on 150 interviews with diverse undergraduate computer science/engineering students, Varma (2010) identifies biases in early socialization and technology-related anxiety as key factors contributing to the underrepresentation of women in the field. Despite similar exam performance between students who are men and women, the study revealed that women were more inclined to underestimate their performance compared to men (Harrington et al., 2018). This suggests that women tend to have less confidence than men in their performance. Similarly, when it comes to students' participation, Brigham and Porquet-Lupine (2021) identifies a minor gender gap in overall participation scores, with no significant difference in how men's and women's grades benefit from participation points. However, they found variation in participation behavior based on format: Men perform better in public participation, while women excel in anonymous participation. Additionally, the authors note that men's dominance in top scores in public participation reinforces stereotypes about their superior aptitude in computer science.

Berman and Bourne (2015) identified another attitudinal factor contributing to the underrepresentation of women in CS education: Women's internalized societal stereotypes about math ability early on affect their performance, course choices, and career paths. They suggest that positive messages and role models can counter these stereotypes, but also note that evidence of success for women in computing careers is crucial to establishing the credibility of these messages and role models. Similarly, Rezwana and Maher (2023) conducted in-depth interviews with women students in introductory programming courses and found a persistent impact of both a lack of prior programming experience and gender bias, contributing to a negative experience for women in these courses.

2.3. Grit

One of the most recognized theories of grit was developed by Duckworth et al. (2007), who define grit as an individual, non-cognitive trait that connotes "perseverance and passion for long-term goals" (2007, p. 1087). Distinguishable from well-known philosophical concepts such as continence and willpower, grit is not merely the capacity to endure the challenges of effort and setbacks or to resist the allure of more immediate rewards. More precisely, the concept of grit assumes the presence of "epistemic resilience" when faced with obstacles that indicate that the likelihood of success is uncertain (Morton & Paul, 2019).

Several recent studies have raised concerns about the validity of the concept of grit. Notably, Crede et al.'s meta-analytic review (Credé et al., 2017) and Crede's critical assessment of grit (Credé, 2018) have been at the forefront of this critique. One of the primary criticisms is that grit closely overlaps with conscientiousness, a key trait within the Big Five personality model (Credé, 2018; Credé et al., 2017). Duckworth et al. (2007) acknowledge this overlap but distinguish grit by emphasizing that "Grit overlaps with achievement aspects of conscientiousness but differs in its emphasis on long-term stamina rather than short-term intensity" (p. 1089). Other criticisms highlight the inconsistent or limited predictive power of grit (Rimfeld et al., 2016; Steinmayr et al., 2018), its unclear theoretical underpinnings, and its limited effectiveness in response to interventions (Credé, 2018). While we acknowledge these criticisms surrounding grit, we recognize that a much larger body of evidence supports its role in predicting success and performance across various domains (Eskreis-Winkler et al., 2014). Researchers have shown grit to be a key factor in professional achievement (Cheung et al., 2021). For instance, a survey of general surgery residents revealed that those with higher levels of grit experienced significantly lower burnout rates and reported improved psychological well-being over six months (Salles et al., 2014). Another study found that employees with higher grit were less prone to burnout, highlighting the potential of cultivating grit as a strategy for organizations aiming to mitigate burnout and enhance employee resilience (Kristinsson et al., 2023).

Research on grit within the field of computer science is scarce (Prickett et al., 2020; Sigurdson & Petersen, 2018), and studies specifically examining gender differences in grit among CS majors are even more limited. In light of this gap, our research contributes to the expanding body of empirical studies on the effect of grit on goal attainment. Particularly, we integrate grit as a contextual variable in the SCCT framework and investigate its relationship with students' intention to graduate from a CS program. Furthermore, given the conflicting evidence in the extant literature on gender differences in grit, we explore whether grit's association with intention varies across gender, addressing a critical gap in the literature.

2.4. Programming experience

Research shows that prior programming experience significantly influences academic performance, engagement, and retention in CS education (Bui et al., 2023; Chen et al., 2019; Fukui et al., 2020; Zheng et al., 2024). Early studies highlight how foundational knowledge in programming fosters a smoother transition into advanced courses. For

example, Fukui et al. (2020) examined the use of binary tree learning systems – educational tools designed to teach students the structure and logic of binary trees, a fundamental data structure in computer science. They found that although students with prior programming experience did not show significantly greater gains in computational thinking compared to those without such experience, the binary tree learning systems still proved effective in supporting learners across varying levels of programming backgrounds. This research suggests that prior experience enhances self-efficacy, even in environments designed for students with diverse programming backgrounds. Additionally, Chen et al. (2019) found that prior experience with graphical programming languages, especially early in education, resulted in more positive attitudes toward programming and better performance. Similarly, Bui et al. (2023) explored how prior programming experience affects academic outcomes in introductory CS courses (CS1 and CS2) and found a consistent performance gap equivalent to a one-letter grade difference between students with prior programming experience and those without. Finally, Zheng et al. (2024) found that students with prior computing experience are more likely to excel in initial CS courses, perform better, and persist in pursuing advanced coursework. The study concluded that factors like programming experience, along with peer comparison and teaching quality, indirectly affect enrollment intentions. Collectively, the studies show that prior programming experience is linked to stronger early performance, more positive attitudes, higher self-efficacy, and indirectly persistence in CS.

The importance of prior programming experience increases as computational thinking becomes essential across disciplines. Courses like the one described by Jung et al. (2017) reflect the growing trend to integrate computational thinking into general education. The demand for practical programming skills has led to interventions like summer boot camps and supplementary courses, which help bridge the gap between high school and college-level programming, particularly for underprepared students (Kirkpatrick & Mayfield, 2017). Bui et al. (2023) highlighted the need for tailored instructional strategies to support students without prior programming experience. Suggestions include providing preparatory courses, boot camps, or supplementary programming tutorials to bridge the knowledge gap before enrolling in CS1. Bowman et al. (2019) investigated how prior programming experience impacts students' pair programming activities. Their study highlights that experienced students often assume leadership roles, guiding the pace and enhancing problem-solving, while novices gain exposure to advanced strategies and build confidence over time. Thus, the study suggests thoughtful pairings and clear role assignments to ensure balanced collaboration and maximized learning for both experienced and novice students.

Recent advances in AI-based tools have had a significant impact on programming education. As Brusilovsky et al. (2023) points out, integrating these tools into learning requires educators to focus more on teaching debugging skills and how to understand code rather than just learning to write it. This shift enables students to engage more deeply with problem-solving, improving their understanding and retention of concepts, regardless of prior experience (Brusilovsky et al., 2023).

Gender differences in prior programming experience also play a role in retention. Bowman et al. (2019) found that women, often with less prior programming experience, face greater challenges in confidence and retention. Bui et al. (2023) emphasized that

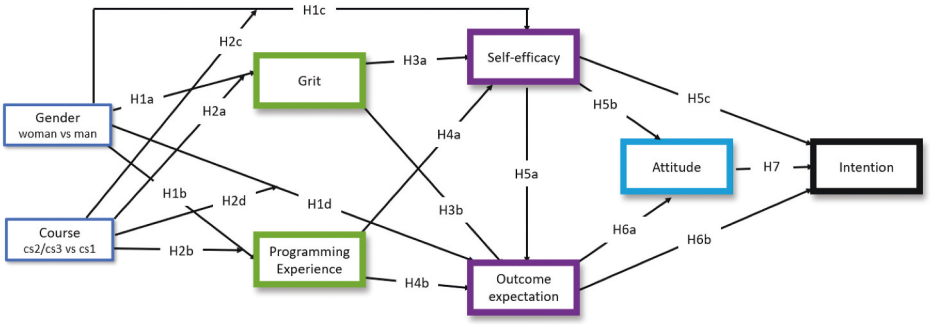


Figure 1. Research model of undergraduate students' intention to complete a computer science degree.

students without prior experience – disproportionately women – struggle more in introductory courses, highlighting the need for targeted educational support to close this gap.

3. Research hypotheses and model

The SCCT framework proposes a complex interplay of social cognitive variables – self-efficacy, outcome expectations, and intention – influenced by personal, contextual, and experiential factors, affecting the self-regulation of behavior (Bandura, 1986; Lent et al., 1994, 2002). This proposition is supported by existing studies (e.g. Lee et al., 2015; Lent, Lopez, et al., 2008; Liguori et al., 2020; Maheshwari et al., 2023; Oben & van Rooyen, 2023; Williams & Subich, 2006), demonstrating that personal and contextual elements, as well as learning experience, shape individuals' self-efficacy beliefs, foster positive outcome expectations, and influence the likelihood of developing intentions specific to a particular domain. In this section, we formulate our research hypotheses based on SCCT and prior work on retention in undergraduate CS education. Our hypotheses are presented as a research model illustrated in Figure 1.

3.1. Individual, contextual, and experiential influences

3.1.1. Gender

Guided by SCCT and existing research, we anticipate that gender will influence self-efficacy and outcome expectations. Research among undergraduate CS students indicates that women generally report lower self-efficacy beliefs than men (Osho et al., 2024). A similar trend is observed among college engineering students, where gender differences in self-efficacy are linked to variations in interests and major choice goals (Inda et al., 2013). Gender also plays a crucial role in shaping outcome expectations. For instance, when it comes to pursuing an entrepreneurial career, as the study by Maes et al. (2014) reveals, men and women differ in their anticipated benefits of pursuing such a career. Women expected that pursuing entrepreneurship would lead to better organization in their lives, allowing them to maintain autonomy and effectively balance the demands of work and family responsibilities. For men, on the other hand, an entrepreneurial career was viewed as a valuable path for attaining wealth and engaging in

a challenging work environment. In computing, Appianing and Van Eck (2015) reported that women exhibited lower expectations for success.

While grit is typically viewed as a relatively stable trait, empirical research also demonstrates that it is malleable (Hwang & Nam, 2021; Ma et al., 2020). Thus, it is reasonable to suggest that grit, particularly in its association with gender, is not an inherently innate trait but is significantly shaped by an individual's environment. Cultural norms and expectations, educational environment, and unique challenges individuals encounter may interact to shape how their grit develops and manifests. In CS programs, for instance, women often encounter systemic obstacles such as widespread stereotypes and a lack of ongoing encouragement and support from family, friends, and faculty, compared to men (Cheryan et al., 2009, 2015, 2017; Cohoon, 2002; Watson, 2020). To navigate and overcome these systemic challenges and barriers, women may develop higher levels of grit, which could help explain why some studies have found that women score higher in grit (e.g. Christensen & Knezek, 2014; Kaya & Yüksel, 2022; Sigmundsson & Leversen, 2024), though some other studies found no gender differences (e.g. Allen, 2014; bin; Shariff et al., 2022; Mohan & Kaur, 2021; Sigmundsson et al., 2021, 2020). Recognizing the dynamic nature of grit opens opportunities for targeted interventions such as exposure to role models and mentorship (bin Shariff et al., 2022) and developing a conducive school culture (Bashant, 2014) to foster grit.

Finally, research reveals significant gender disparities in programming experience, with women generally reporting much less programming experience than men (Beyer et al., 2005; Murphy et al., 2006; Osho et al., 2022). Based on the foregoing, we propose the following hypothesis:

> **Hypothesis 1:** Significant gender differences exist in terms of (a) grit, (b) programming experience, (c) self-efficacy, and (d) outcome expectation.

Further, we posit that course level will act as a moderator of the relationship between gender and grit, programming experience, self-efficacy, and outcome expectations, as gender disparities have been shown to vary over time. For example, research indicates that although women generally begin programming later than men (Beyer et al., 2005; Murphy et al., 2006), they can match men's mastery of programming concepts by the end of their introductory CS courses (Murphy et al., 2006). Building on these findings, we anticipate that as men and women undergraduates advance in their computing coursework, they will develop higher levels of grit, enhance their programming experience, and show growth in self-efficacy and outcome expectations. However, the pace and magnitude of these improvements will differ by gender. Accordingly, we propose the following hypothesis:

> **Hypothesis 2:** Course level moderates the relationship between gender and (a) grit, (b) programming experience, (c) self-efficacy, and (d) outcome expectation.

3.1.2. Grit

In this work, grit refers to perseverance for completing a CS degree. In educational contexts, the impact of grit on success has produced mixed results, with some studies identifying grit as a predictor of various success outcomes, such as academic achievement, retention, grade point average, and even performance in events like the National Spelling Bee (Duckworth et al., 2007). However, other studies on high-school computer

programming (e.g. Thanasuan et al., 2017) and first-year undergraduate Computer Science (e.g. Prickett et al., 2020; Sigurdson & Petersen, 2018) suggest that grit may not serve as a reliable predictor of success. While both elements – perseverance and passion – of grit may contribute to predicting academic success, perseverance has consistently emerged as the stronger and more positive predictor (Christopoulou et al., 2018; Muenks et al., 2018).

Due to the inconsistent predictive power of grit, Credé (2018) recommends investigating how grit interacts with other traits to better predict success or performance. Researchers examining the influence of grit alongside other traits have consistently reported positive outcomes. For instance, grit has been shown to positively correlate with self-efficacy, which partially or fully mediates the relationship between grit and academic outcomes in early adolescent students (Usher et al., 2019). Similarly, Alhadabi and Karpinski Alhadabi and Karpinski (2020) found that grit is positively linked to academic performance through a sequential chain of mediators, including self-efficacy and achievement-oriented goals, further reinforcing the importance of studying grit in conjunction with other psychological factors. Additionally, research conducted among students at an urban middle school in New Jersey found that those with stronger outcome expectations – specifically personal and social normative expectations – were more likely to display higher levels of grit (Verdesco, 2016). Consequently, we hypothesize the following:

> **Hypothesis 3:** Students with more grit exhibit higher levels of (a) self-efficacy and (b) outcome expectation.

3.1.3. Programming experience

In our study, we operationalize programming experience as students' self-assessed programming skills relative to their course requirements. According to SCCT, learning experiences – particularly those arising from direct performance accomplishments – strongly influence an individual's self-efficacy and outcome expectations (Lent et al., 1994). Programming courses are typically required for CS majors (Pioro, 2004), and many computing-related jobs demand programming experience.[1] Consequently, when students perceive that they have the skills required to complete their programming classes, it reinforces both their confidence to master their degree and their belief in achieving favorable outcomes, such as securing a well-paying job in the computing field. Therefore, we hypothesize as follows:

> **Hypothesis 4:** Students with more programming experience exhibit higher levels of (a) self-efficacy and (b) outcome expectation.

3.2. Self-efficacy, outcome expectation, and attitude

3.2.1. Self-efficacy

In the context of our study, self-efficacy refers to students' domain-specific confidence in their ability to succeed in computer science, including understanding coursework, performing well on labs and projects, solving difficult programming problems, handling unexpected challenges, and completing a CS major. Within the SCCT framework (Lent et al., 1994; Lent, Lopez, et al., 2008), self-efficacy is understood to be a critical mechanism that shapes both outcome expectations and the formation of intentions. When

individuals perceive themselves as competent in a given activity, they are more inclined to anticipate favorable results from engaging in that activity (Lent et al., 1994). In the computing discipline, such positive beliefs about one's abilities have been linked to enhanced outcome expectations and stronger intentions to persist in the field (Lent et al., 2011; Lent, Lopez, et al., 2008). Moreover, evidence from other domains, such as entrepreneurship (Liguori et al., 2020) and computer-supported education (Yesilyurt et al., 2016), further underscores self-efficacy as a key predictor of one's attitude toward the discipline. Therefore, we posit that:

> **Hypothesis 5:** Students' self-efficacy is positively associated with (a) outcome expectation, (b) attitude toward CS, and (c) intention to complete their CS program.

3.2.2. Outcome expectation

In alignment with SCCT (Lent et al., 1994, 2002), we define outcome expectations as students' beliefs about the likely benefits and consequences of completing an undergraduate CS degree. These include intrinsic expectations, such as job satisfaction and personal achievement, and extrinsic expectations, including high earning potential, status, and career prestige (Lent et al., 1994; Leventhal & Chilson, 1989). The SCCT framework highlights that outcome expectations can directly impact intentions to pursue specific academic goals (Lent et al., 1994). Research shows that students' outcome expectations predict their goal to complete a STEM degree (Byars-Winston et al., 2010). We also anticipate this effect for computer science: Students who strongly believe that earning a CS degree will lead to desirable outcomes – such as securing a well-paying job or achieving personal growth – would be more likely to sustain their commitment and actively strive to graduate.

Existing studies further demonstrate a significant association between outcome expectations and attitudes. Synonymous with behavioral beliefs (Bohlen et al., 2022; Compeau & Higgins, 1995) in the Theory of Planned Behavior (Ajzen, 1991), outcome expectations, when denoted as perceived consequences, are strongly linked to an individual's attitude toward a behavior (Ajzen, 2020). Liguori et al. (2020) showed that individuals who anticipate more favorable outcomes from entrepreneurship generally display stronger entrepreneurial attitudes. Similarly, Han (2017) observed that positive public attitudes toward science and technology correlate with heightened student expectations of pursuing STEM careers. We hypothesize similar effects in the CS field:

> **Hypothesis 6:** Students' outcome expectation is positively associated with their (a) attitude toward CS and (b) intention to complete their CS program.

3.2.3. Attitude toward CS

Students' attitude toward CS include their affective (emotional feelings toward CS), evaluative (thoughts, beliefs, and judgment about CS), as well as behavioral (how they act or behave in response to the CS discipline) attitudes (Breckler & Wiggins, 1989; Netzer et al., 2018). Strong evidence exists that an individual's attitude influences their intentions (Ajzen, 1991). In a community college study, women who reported a strong intention to pursue computing and information sciences at a four-year university held positive attitudes toward computer programming and placed high value on computing (Denner et al., 2015). Therefore, we hypothesize as follows:

Hypothesis 7: Students' attitude toward CS is positively associated with their intention to complete their CS program.

4. Positionality statement

The survey study data presented in this paper are part of a multi-year project to study and improve the retention and graduation rates of underrepresented students in the CS programs at two Historically Black Colleges and Universities (HBCUs) and one Predominantly White Institution (PWI).[2] The researchers are not the instructors of the courses studied in this project, allowing us to maintain a certain level of objectivity in assessing these courses. To get a better understanding of the classroom environment, instructors are interviewed at the end of each semester.

The project team consists of students and faculty at the three institutions under investigation, plus two external consultants. Whereas the senior members of our team contribute their extensive experience in CS education research, upper-division undergraduate students are purposefully involved in the project since they have more recent personal experiences with the studied courses. Collectively, the team members span a wide variety of genders and races, including foreign nationals who did not complete their undergraduate degrees at a US university.

Our team takes a social constructionist approach to race and gender but at the same time acknowledges that a reductionist, label-based approach is often unavoidable in large-scale survey data collection and subsequent quantitative analyses. We rely on the nuanced, intersectional experiences of the team members to critically assess and contextualize the outcomes of our research.

5. Research methods

5.1. Participants and procedures

In Spring 2023, a total of 584 undergraduate students enrolled in various sections of CS1 (Introduction to Programming), CS2 (Programming Methodology), and CS3 (Algorithms and Data Structures) at a large, engineering-focused, predominantly white institution (PWI) in the southeastern United States, participated in the study in the Spring of 2023. The study procedures were reviewed and approved by our Institutional Review Board (IRB). The recruitment process involved inviting the instructors of these courses to participate in research aimed at broadening participation. Once they agreed, visits were made to their classes to introduce the project and ask students to participate in a research survey for extra credit in their respective courses. Those who did not wish to participate in the research were given the option to earn extra credit by completing a non-research survey. The study was conducted using Qualtrics.

Of the invited students, 459 participated in the survey (representing 78.6% response rate). Data from 9 participants were removed because they failed the attention question. An additional 147 participants were removed because they were not enrolled in the CS major. Of the remaining 303 participants, 243 (80.20%) were men, 55 (18.15%) were women, 4 (1.32%) identified as non-binary, and 1 (0.33%) respondent selected "Prefer not to answer". Due to the few respondents who selected the last two gender options, we considered only the data from students who reported their gender as man or woman (i.e.

298 participants) in our statistical models. We do, however, present comparative descriptive results of the non-binary participants in Section 6.6. In terms of course level, 70 (23.10%), 135 (44.55%), and 98 (32.34%) were enrolled in CS1, CS2, and CS3 courses, respectively. Regarding race, 2 (0.66%) identified as American Indian; 42 (13.86%) as Asian; 21 (6.93%) as Black or African American; 1 (0.33%) as Pacific Islander; 221 (72.94%) as White; 11 (3.63%) as multiracial; and 5 (1.65%) selected "Other". The predominance of white participants prevented us from including race as a parameter in our statistical models. Still, descriptive results comparing Black/African American and white students, as well as results across both race and gender, are presented in Section 6.6.

5.2. Measures

The first part of our survey introduced participants to the study's purpose and presented the informed consent document. The next section contained demographic questions and their current course level and section. Participants were then asked about their subjective experiences (see Appendix), including grit, programming experience, self-efficacy, attitude toward the computing discipline, and intention to complete their CS degree. The subjective experience questions were adopted from Osho et al. (2024), which comprehensively tested their validity and robustness. All subjective experience scales, except for the intention scale, were measured using a 5-point Likert scale ranging from disagree (−2) to agree (2). For the intention scale, participants rated how likely they were, from unlikely (−2) to likely (2), to pursue the specified activities or goals.

Grit was assessed using a 5-item scale that assessed participants' perseverance for completing a CS degree by capturing students' perseverance-related self-regulatory tendencies at the time of the survey. The items reflected core facets of grit, including diligence and sustained effort, as well as responses to obstacles, with reverse-coded statements capturing procrastination and giving up. For example, participants responded to statements such as "I can overcome obstacles to complete my tasks", which reflects persistence through difficulty, and "I give up when problems seem too difficult", which inversely gauges the tendency to abandon challenging endeavors. Accordingly, we interpret this measure as a contemporaneous indicator of students' persistence tendencies, rather than as an immutable trait.

Programming experience was evaluated by examining students' perceived experience relative to the demands of the focal course. To measure this, participants were asked to respond to 2 specific questions that assessed how their programming experience matched their current course demands (e.g. "The course work in my current course required less programming experience than I had".) By capturing these responses, the study aimed to understand how participants' perceived programming skills matched the expectations and content of their current courses, providing insight into the adequacy and relevance of their programming experience.

The self-efficacy scale assessed how confident participants felt about their capability as a Computer Science student to fulfill their CS course requirements and complete a CS degree. This scale consisted of 8 questions that captured various aspects of self-assurance related to computer science tasks and academic requirements. For instance, participants responded to statements like, "I am confident I will do well in computer science labs and projects", which directly measured their belief in their competence in practical CS tasks.

Conversely, "I'm not good at computing" was included to gauge lower self-efficacy, and responses to this item would indicate a lack of confidence. The items allowed for a nuanced understanding of participants' self-perceived abilities in their CS academic journey.

Outcome expectation was measured by presenting participants with 8 potential outcomes associated with earning a degree in a computing discipline (e.g. "I expect that learning to use computer science skills will help me achieve my career goals" and "I see computer science as a subject I will rarely use in my daily life"). These items were designed to capture the range of expectations participants might have about the practical benefits and relevance of computer science in their future careers and daily lives.

Attitude toward computer science (CS) was measured with a 5-item scale, which aimed to capture participants' overall attitude and general disposition toward CS. This scale included statements designed to reflect both positive and negative attitudes toward CS (e.g. "Computer science is a worthwhile and necessary subject" and "I do not enjoy spending a lot of time writing programs".)

The Intention to complete a CS degree was evaluated through a 6-item scale to gauge participants' commitment to follow through with their academic plans in CS. Participants were presented with positive and negative statements and asked to express how likely they were to engage in the related activities or achieve the specified goals (e.g. "How likely are you to complete a CS major or minor?" and "How likely are you to change to a different major or minor?"). This approach allowed us to capture participants' intentions regarding their education in CS, providing insight into their level of determination and continued commitment to completing their CS degree.

5.3. Data analysis

To explore if grit and experience differ by gender and course level and whether such differences subsequently cause differences in student self-efficacy, outcome expectation, attitude, and intention to complete their computer science degree, we developed a mediated regression model using path modeling. The model used gender, course level, and their interaction as exogenous variables and had two mediators: grit and experience. Path modeling is a robust analytical tool for examining complex associations and for specifying and estimating hypothesized direct and indirect relationships among variables (Lleras, 2005).

In the path analysis, we treated gender as a categorical variable, with men as the reference (baseline) group. The course level was modeled as a three-level categorical predictor (CS1, CS2, CS3) using two dummy variables (cs2 and cs3), with CS1 serving as the reference category. The subjective experience constructs were determined using composite scores. Path analysis with composite scores is a commonly employed method in data analysis, favored for its effectiveness in predicting and diagnosing individual outcomes (Deng & Yuan, 2023). We used Cronbach's alpha (CA) to measure the internal consistency and reliability of the constructs and ensure the items within a construct measure the same underlying concept. A high Cronbach's alpha value indicates that the items are strongly correlated and provide a reliable measure of the construct (Streiner, 2003). For each construct, we excluded items one at a time until a maximum CA value was obtained. By pinpointing items that negatively impact the overall CA, we eliminated those items to

Table 1. Means, standard deviations, correlations, and Cronbach's alphas of variables.

	Mean	SD	1	2	3	4	5	6
1. Grit	1.27	0.65	**0.81**					
2. Programming experience	−0.55	1.13	0.18**	**0.70**				
3. Self-efficacy	0.99	0.78	0.55***	0.47***	**0.91**			
4. Outcome expectation	1.64	0.50	0.39***	0.15*	0.46***	**0.86**		
5. Attitude	1.26	0.69	0.43***	0.20**	0.49***	0.66***	**0.71**	
6. Intention	1.20	0.58	0.39***	0.21***	0.48***	0.66***	0.58***	**0.73**

Note: Diagonal values are Cronbach's alpha. Correlation significance: ***$p < .001$, **$p < .01$, *$p < .05$.

strengthen the internal consistency of the scale (Pereira da Hora et al., 2014). Table 1 presents Cronbach's alpha for the scales. To determine the relationships among the scales, we calculated the average scores of the items for each scale, reversing all negatively phrased items before computing the average. Averaging the scale scores allows for easier comparisons among scales with varying numbers of items. It also preserves the original scale metric of the items, making the results more interpretable (DiStefano et al., 2019). For differences of the continuous variables between groups, i.e. gender and course levels, we present standardized effects as a Cohen's *d*, and for associations between two continuous variables, we present the standardized effect using a beta coefficient. The analysis was performed using the lavaan package in R, with the model fitted using the Maximum Likelihood with Robust Standard Errors (MLR) estimator.

6. Results

This section describes the result of our path model analysis. To fit our model, we adopted an exploratory, saturated-model approach, which allows us to consider and explore all possible connections among the sources of intention to graduate from a CS program and how these sources differ by gender and course level. Table 1 provides the means, standard deviations, intercorrelations, and Cronbach's alphas for the scales. After fitting the model with a saturated path model, non-significant effects were iteratively trimmed from the model until only significant paths remained. The final model is presented in Figure 2. The model has a good[3] fit: $\chi^2(25) = 32.518$, $p = .144$; RMSEA = 0.032, 90% CI: [0.000, 0.060]; CFI = 0.989; and TLI = 0.981. In addition to the path model, Figure 3 demonstrates the combined effect of the various direct and mediated pathways by means of the marginal effects of gender and course level on each subjective experience construct.

6.1. Gender and course level differences in grit and programming experience

We first examine the associations of gender and course level with students' perseverance toward completing a CS degree (i.e. grit). Our path model (Figure 2) indicates a significant association between gender and grit (H1a), and this association is moderated by course level (H2a) (interaction effect: $\chi^2(2) = 9.201$, $p = .010$). The moderation effect is represented graphically in the top orange box in Figure 2. In CS1, women show a significantly higher level of grit than men ($d = 0.429$, $p = .007$). The interaction effect shows that the difference becomes slightly less strong in CS2 (a decrease of $d = -0.012$ compared to CS1, for a total difference of $d = 0.417$), but in CS3, women actually show a substantially *lower* level of grit than men (a decrease of $d = -0.689$ compared to CS1, for a total

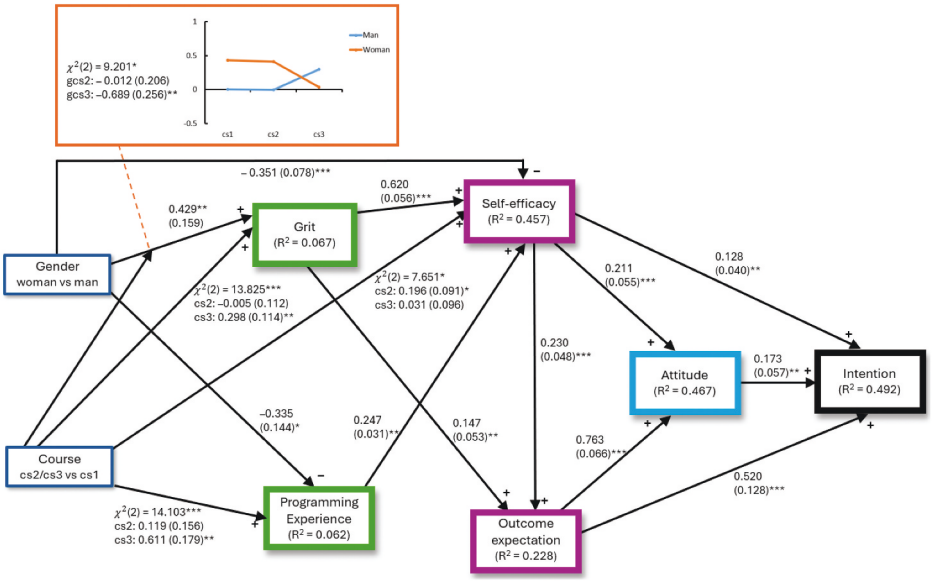


Figure 2. The path model. Labels on arrows represent standardized regression coefficients, standard errors (in parentheses), and significance: ***$p < .001$, **$p < .01$, *$p < .05$. Moderation effect statistics and graphical representation are presented in the orange box.

difference of $d = -0.272$). For men, grit is significantly higher in CS3 compared to CS1 ($d = .298$, $p = .009$). The marginal effects graph of gender and course level on grit (Figure 3(a)) further illustrates these relationships.

Next, we examine the associations of gender and course level with students' perceived programming experience relative to course requirements (i.e. programming experience). Figure 3(b) demonstrates these relationships: Confirming H1b, women report significantly less programming experience (i.e. they feel less prepared for the course) than men ($d = -0.335$, $p = .020$). A significant association (although not hypothesized) of course level with programming experience is found ($\chi^2(2) = 14.103$, $p < .001$): while no difference between CS1 and CS2 exists, students in CS3 exhibit higher programming experience than those in CS1 ($d = 0.611$, $p = .001$). Finally, there was no significant gender and course-level interaction for programming experience (H2b not supported).

6.2. Self-efficacy

We now evaluate how confident participants felt about their capability to succeed in CS (i.e. their self-efficacy). Confirming H1c, women report significantly lower self-efficacy ($d = -0.351$, $p < .001$) than men. Self-efficacy also significantly differs by course level ($\chi^2(2) = 7.651$, $p = .022$). Students in CS2 have higher self-efficacy than CS1 students ($d = 0.196$, $p = .030$). However, the gender × course-level interaction for self-efficacy was not significant (H2c not supported).

Both grit (H3a) and programming experience (H4a) are significantly and positively associated with students' self-efficacy ($\beta = 0.620$, $p < .001$ and $\beta = 0.247$, $p < .001$, respectively). Because gender and course level are associated with grit and

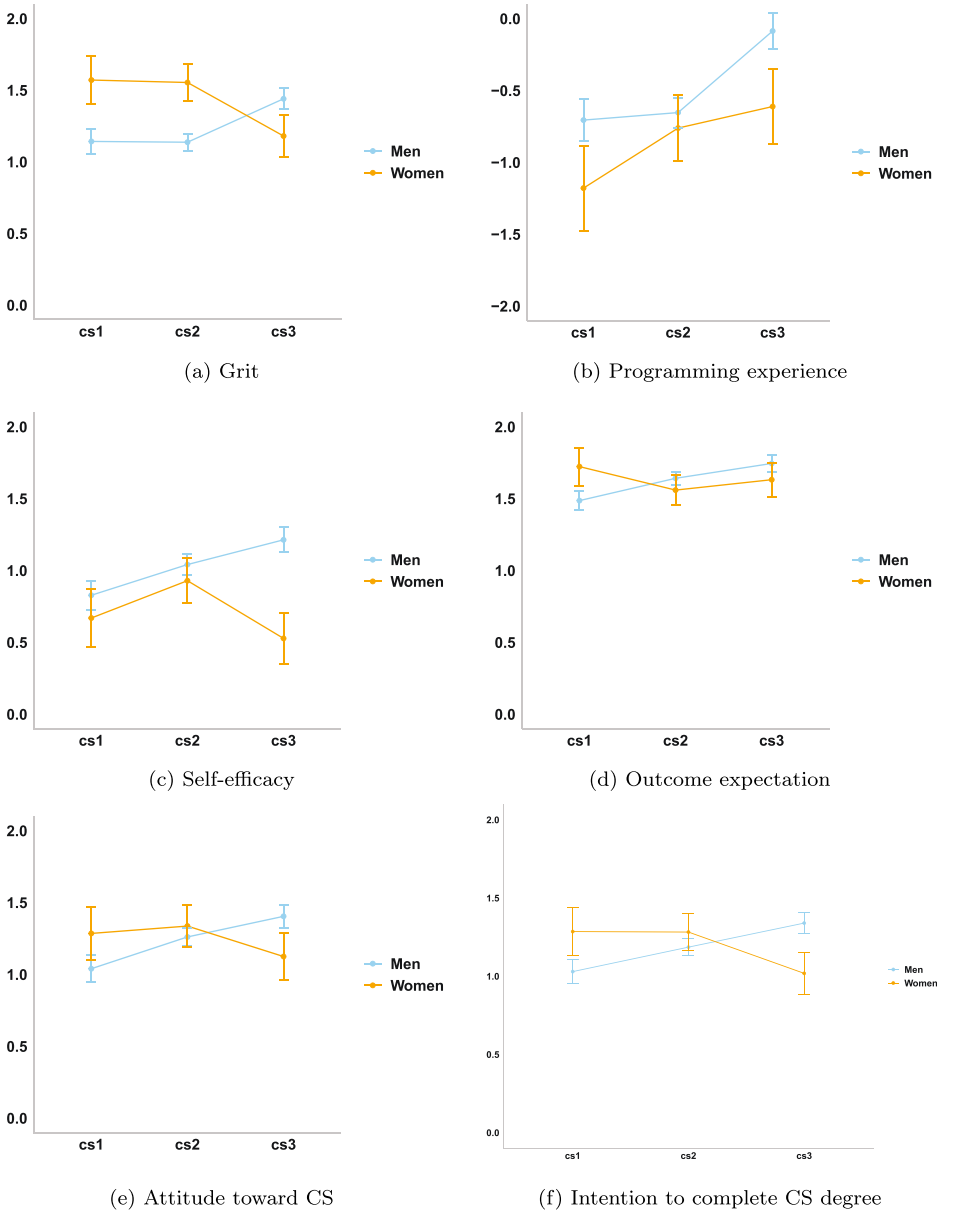


Figure 3. Marginal effects of gender and course level on the factors in our research model. Because the factors are scale-free latent variables, their values are fixed to zero for men in CS1. The y-axis is scaled by the sample standard deviation.

programming experience, these constructs partly account for the association between gender/course level and self-efficacy (i.e. a pattern consistent with partial statistical mediation). Figure 3(c) presents the marginal associations of gender and course level with self-efficacy, combining direct and indirect components. Women show slightly lower self-efficacy than men in CS1 and CS2; however, the gap widens

in CS3. Whereas men's self-efficacy increases with each course level, women's self-efficacy plateaus at CS2.

6.3. Outcome expectation

In terms of students' expected benefits from completing a CS degree (i.e. outcome expectation), we find that gender was not significantly associated with outcome expectation (H1d not supported). However, while grit (H3b) shows a positive association with outcome expectations ($\beta = 0.147$, $p = .006$), programming experience (H4b) did not. Self-efficacy (H5a) was positively associated with outcome expectations ($\beta = 0.230$, $p < .001$) and statistically accounted for portions of the grit–outcome and programming experience–outcome associations (consistent with partial and full statistical mediation, respectively). In turn, grit, programming experience, and self-efficacy collectively accounted for the gender and course-level associations with outcome expectations. Figure 3(d) displays the marginal associations of gender and course level with outcome expectations, combining direct and indirect components. It shows an increase in outcome expectations for men as they progress from CS1 to CS3. For women, there was a decrease in outcome expectation from CS1 to CS2, and a slight increase thereafter. Women start higher in CS1, but men exceed them in CS2 and CS3.

6.4. Attitude toward CS

Students' self-efficacy (H5b) and outcome expectations (H6a) show a significant positive association with their overall attitude and general disposition toward CS ($\beta = 0.211$, $p < .001$ and $\beta = 0.763$, $p < .001$, respectively). In addition to its direct association, the self-efficacy-attitude link appears to be partly accounted for by outcome expectations. Together, self-efficacy and outcome expectations statistically account for the relationships of grit and programming experience, as well as gender and course level, with attitudes toward CS. Figure 3(e) depicts these indirect pathways for gender and course level. Attitudes are higher in CS2 than CS1 for both groups, with a larger increase for men. By CS3, men's attitudes continue to rise, whereas women's decline. This decrease in women's enthusiasm toward CS at higher course levels may be an early indicator of reduced retention rates.

6.5. Intention to complete CS degree

Finally, we evaluate students' commitment to follow through with their academic plans in CS (i.e. their intention to complete their CS degree). Students' self-efficacy has a significant positive relationship with their intention to complete their CS degree (H5c, $\beta = 0.128$, $p = .001$). In the same vein, both students' outcome expectations (H6b) and their attitudes toward CS (H7) were positively associated with their intention to complete their CS degree ($\beta = 0.520$, $p < .001$ and $\beta = 0.173$, $p = .002$, respectively). Given the observed associations of self-efficacy with both of these variables, we can argue that its association with intention to complete a CS degree is partially mediated.

The associations of gender and course level with intention appear to be fully accounted for by upstream constructs in the model (i.e. attitude, outcome expectations,

Table 2. Group means and standard deviations (in parentheses) of subjective experience by gender.

Gender	N	Grit	Prog. experience	Self-efficacy	Outcome exp'n	Attitude	Intention
Man	243	1.23 (0.64)	−0.49 (1.17)	1.05 (0.76)	1.64 (0.49)	1.26 (0.66)	1.20 (0.57)
Woman	55	1.44 (0.67)	−0.82 (0.93)	0.73 (0.82)	1.62 (0.56)	1.25 (0.83)	1.20 (0.64)
Non-Binary	4	1.31 (0.55)	−0.88 (1.03)	0.62 (0.95)	1.41 (0.70)	1.12 (0.63)	0.67 (0.97)

Table 3. Group means and standard deviations (in parentheses) of subjective experience by race.

Race	N	Grit	Prog. exp.	Self-efficacy	Outcome exp'n	Attitude	Intention
American Indian	2	1.12 (1.24)	−1.00 (0.71)	1.06 (1.15)	1.88 (0.18)	1.12 (1.24)	1.42 (0.35)
Asian	42	0.99 (0.71)	−0.75 (0.78)	0.76 (0.64)	1.50 (0.47)	1.12 (0.66)	1.13 (0.62)
Black	21	1.29 (0.58)	−0.93 (0.99)	0.76 (0.74)	1.34 (0.86)	0.90 (0.88)	1.09 (0.63)
Pacific Islander	1	2.00 (-)	0.00 (-)	1.62 (-)	1.50 (-)	1.25 (-)	1.67 (-)
White	221	1.32 (0.63)	−0.48 (1.20)	1.06 (0.78)	1.69 (0.44)	1.32 (0.67)	1.21 (0.58)
Multiracial	11	1.16 (0.53)	−0.68 (1.08)	0.67 (1.00)	1.52 (0.50)	1.05 (0.70)	1.14 (0.41)
Other	5	1.50 (0.98)	−0.10 (1.14)	1.00 (1.00)	1.35 (0.85)	1.15 (0.76)	1.10 (0.93)

self-efficacy, programming experience, and grit). Combining direct and indirect components, Figure 3(f) shows a pattern similar to attitudes toward CS: women begin CS1 with higher intention than men; both groups rise in CS2, with a larger gain for men; by CS3, men's intention increases further, whereas women's intention declines.

6.6. Descriptive comparisons across gender, race, and intersectional groups

One of the drawbacks of this study is the small number of non-binary and Black participants. As a result, the sample does not support reliable statistical comparisons between binary and non-binary students, Black and non-Black students, or their intersectional groups. Consequently, our inferential analyses excluded non-binary participants and did not include race. However, we report descriptive statistics for these groups, situating their values alongside those of the majority participants. Tables 2–4 present means and standard deviations for the subjective-experience constructs by gender, race, and the intersection of race and gender.

In terms of gender, as shown in Table 2, non-binary participants report levels of grit that fall between those of women and men, but they report the lowest level of programming experience. They also exhibit the lowest self-efficacy, outcome expectations, attitudes toward CS, and intention to complete the degree, compared with both women and men. Table 3 shows that Black students report the lowest outcome expectation, attitudes toward CS, and intention to complete the degree, compared with students of other races. When we combined race and gender (Table 4), we observe that non-Black women show lower levels of programming experience but higher levels of grit, self-efficacy, outcome

Table 4. Group means and standard deviations (in parentheses) of subjective experience by race and gender.

Race	Gender	N	Grit	Prog. exp.	Self-efficacy	Outcome exp'n	Attitude	Intention
Black	Man	15	1.25 (0.45)	−1.00 (1.09)	0.81 (0.64)	1.36 (0.66)	0.95 (0.74)	1.08 (0.51)
Black	Woman	6	1.38 (0.88)	−0.75 (0.76)	0.62 (1.02)	1.29 (1.32)	0.79 (1.24)	1.11 (0.92)
Non-Black	Man	228	1.23 (0.65)	−0.45 (1.17)	1.06 (0.76)	1.66 (0.47)	1.28 (0.65)	1.21 (0.57)
Non-Black	Woman	49	1.44 (0.65)	−0.83 (0.96)	0.74 (0.80)	1.67 (0.40)	1.31 (0.77)	1.21 (0.61)

expectations, attitudes, and intention than Black women, who on several outcome variables seem doubly affected by their intersectional identity.

7. Discussion

7.1. Summary of findings and theoretical implications

This study explored how gender differences in grit and programming experience across introductory computing course levels influence undergraduate computer science (CS) students' intention to complete their CS degree. We developed a research model based on social cognitive career theory (SCCT). SCCT proposes that the self-regulation of career-related behaviors comprises a complex interplay of social cognitive variables – self-efficacy, outcome expectations, and goal representations – influenced by learning experiences and person and contextual factors (Bandura, 1986; Lent et al., 1994, 2002). In the context of our study, the framework suggests that self-efficacy and outcome expectations will have a direct association with students' intention to graduate from their CS programs. Both self-efficacy and outcome expectation are associated with grit and programming experience, which, in turn, are related to individual and contextual variables, viz., gender and course level. Drawing from other existing studies (Liguori et al., 2020), we incorporated attitude in SCCT as an important mediator of the relationship between self-efficacy and outcome expectations on the one hand and intention on the other. Our study offers empirical support for applying SCCT to understand the factors that shape undergraduates' intentions to complete their CS degree.

The findings of this study offer several insights into how gender differences in grit and programming experience, along with students' progression through various course levels, relate to their intentions to complete a CS degree. First, our results confirm that women and men differ significantly in grit, perceived programming experience relative to course requirements, and self-efficacy, and that these differences vary by course level. Specifically, women showed higher levels of grit than men in CS1 and (less substantially) in CS2, but not in CS3; women had lower levels of perceived programming experience and self-efficacy than men at all course levels. Combined, these associations result in the observed gender differences in outcome expectations, attitude toward CS, and degree completion intention – all of which show more substantial increases across course levels for men than for women. For instance, regarding outcome expectations associated with earning a CS degree, although women start higher in CS1, men exceed them in CS2 and CS3. For women, outcome expectations drop from CS1 to CS2 and then recover slightly. For men, outcome expectations rise steadily from CS1 to CS3. While gender differences in self-efficacy, outcome expectation, attitude, and intention have been studied before (Huang, 2013; Osho et al., 2024) (and indeed, our results align with previous findings on gender differences in self-efficacy (Lishinski et al., 2016)), the grit- and experience-based explanation for these observed gender differences is a key contribution of this paper.

Our study does not investigate the cause of the fluctuating gender differences in grit. Still, we speculate that women may receive a boost in motivation after passing their first programming course (post-CS1), only to lose it as they progress further through their degree program. The initial boost in motivation can be explained by existing research indicating that women often attribute their academic achievements

to their hard work and find motivation in academic incentives, while men tend to associate their success with their innate abilities (Christopoulou et al., 2018). One might argue that the subsequent "loss" of grit is due to the stereotypes (Berman & Bourne, 2015) and biases (Murphy et al., 2006; Rezwana & Maher, 2023) "catching up" (i.e. accumulating over time and taking a toll) with women students around the time they take the algorithms and data structures course (CS3). The study of gender differences in the context of computing education is an evolving area. As such, further in-depth research is needed to thoroughly understand the root causes contributing to the variations in gender differences in grit observed across different course levels.

Similarly, while our study does not uncover the exact reason for women's lack of perceived programming experience, one possible explanation for this deficit is that men tend to have more prior exposure to computer programming and are generally more interested in computer programming before they start college (Du & Wimmer, 2019; Rezwana & Maher, 2023). Another possibility is that women underestimate their abilities (Harrington et al., 2018).

Grit positively predicts self-efficacy and outcome expectations. Programming experience exhibits a direct positive association with self-efficacy, but not with outcome expectations. Instead, self-efficacy appears to serve as a crucial mediating mechanism. These findings suggest that strengthening students' grit may coincide with higher confidence in their abilities and more favorable expectations about the benefits of studying computer science, which together align with more positive attitudes toward the field and stronger intentions to complete the degree. Likewise, programming experiences appear most beneficial when they build self-efficacy because increased confidence is then linked with more favorable outcome expectations. These factors ultimately reinforce their commitment to completing a CS degree. Thus, while baseline differences in grit and perceived programming experience matter, the paths by which they connect with broader motivational constructs and career intentions are complex and are routed through self-efficacy and outcome expectation.

From a theoretical perspective, augmenting SCCT with grit, programming experience, and attitude increases its explanatory precision for CS persistence. Programming experience operationalizes learning experiences in a discipline-specific way, clarifying how gender and course level can shape self-efficacy and outcome expectations. Grit adds a self-regulatory person factor that helps explain differences in sustained effort and the accumulation and interpretation of mastery experiences. Attitude introduces a distinct evaluative pathway, allowing the model to differentiate whether intention is shaped primarily by self-efficacy, outcome expectations, and/or attitude.

The expanded SCCT model offers a reusable template for theory testing in CS and similar domains. It specifies where experience accumulation and self-regulation enter the SCCT chain and how beliefs translate into intention via attitudinal appraisal, enabling clearer cross-study comparisons and stronger mediation-based explanations. It also provides a coherent structure for integrating contextual supports/barriers as upstream predictors of experience and beliefs. Researchers can use the model to compare whether disparities emerge primarily from unequal access to experience-building opportunities, differences in how accurately people judge their own ability, divergent expectations about the value of the degree, or differences in self-regulation under challenge.

7.1.1. Difference in the subjective experience across gender, race, and intersectional groups

Our descriptive contrasts point to systematically less favorable subjective experiences among gender-minority students and persistent disparities by race. Non-binary participants report the lowest programming experience, self-efficacy, outcome expectations, attitudes toward CS, and intention relative to women and men. By race, Black students score lower than non-Black students in outcome expectation, attitudes toward CS, and intention to complete the degree. Considering race and gender together, Black women exhibit lower grit, self-efficacy, outcome expectations, attitudes, and intention than non-Black women. These findings are consistent with evidence of unequal or exclusionary climates for underrepresented or marginalized groups, including Black, Women of color, non-binary/gender-minority/gender-nonconforming students in STEM and CS education, and the downstream links to confidence, expectations, and persistence intentions (Boman et al., 2024; Du & Wimmer, 2019; Hunt et al., 2022; Osho et al., 2022, 2024; Skorodinsky, 2024; Sloughter et al., 2019).

Caveat: the numbers of non-binary and Black participants in our sample are small; therefore, these subgroup contrasts are descriptive and should not be treated as reliable statistical comparisons between binary and non-binary students, Black and non-Black students, or their intersections.

7.2. Implications for practice

Our findings demonstrate that a lack of programming experience is partially to blame for the gender deficits in self-efficacy, outcome expectation, attitude, and intention. In a practical sense, this finding suggests that increasing girls' exposure to meaningful computer science experiences before college may help reduce these deficits. This interpretation is consistent with prior evidence showing that early, targeted programming opportunities can narrow gender gaps in motivation and confidence. Girls who received programming exposure reported greater technology interest and self-efficacy than girls who did not, and their interest and self-efficacy were comparable to boys', with no significant gender difference (Allaire-Duquette et al., 2022; Master et al., 2017).

However, given the pervasive influence of societal stereotypes and biases, girls may be less inclined to seek out these opportunities on their own. Consequently, it becomes vital for educators to incorporate such experiences into classroom settings, ideally in ways that resonate with a wide range of learners and relate to real-world contexts (e.g. B Daily et al., 2015). Adapting DuBow et al'.s (2016) Engagement Practices Framework, we recommend structuring introductory programming courses around purposeful relevance, inclusive community building, and confidence/identity development. Instructors can make programming matter by using projects and examples grounded in students' everyday lives (e.g. data from sports, music, social media, campus life, or community issues), explicitly connecting programming to other disciplines (e.g. biology, business, art, cybersecurity), and directly addressing misconceptions (e.g. that programming requires "genius" talent or extensive prior experience), while offering meaningful project choice when feasible. Beyond relevance, translating access into sustained participation requires a supportive classroom culture. Educators can grow a positive programming community by establishing welcoming norms for labs and code reviews, discouraging stereotypes about who

"belongs" in programming, and using structured collaborative learning (clear roles, accountability, and rotation), including grouping or scaffolding teamwork by prior experience so novices can participate equitably. Finally, because confidence is built through repeated, supported engagement, instructors can strengthen students' programming self-efficacy and professional identity by emphasizing a growth mindset about debugging and learning, providing timely feedback focused on improvement (e.g. incremental milestones and revision opportunities), creating frequent points of contact through guided lab time and accessible office hours, and mitigating stereotype threat through inclusive language and visible, diverse role models.

We further note that our programming-experience measure is subjective and course-referenced, capturing perceived experience relative to course requirements. Thus, women's lower ratings could reflect underestimation of preparedness (or stricter self-evaluation standards) rather than a true deficit in objective programming experience. Prior research shows that women are less likely than men to rate their ability favorably relative to classmates and tend to hold higher personal performance standards (Hunt et al., 2022). This perspective calls for confidence-building exercises, which may be crucial to the success of the women students, as well as the non-women students, who may underestimate their abilities. Furthermore, those who routinely engage with these students, including instructors, should recognize that a student's seeming lack of confidence may not accurately reflect their actual competence. Therefore, they must take proactive steps to validate and encourage such students' abilities rather than misinterpret self-doubt as evidence of limited potential. However, such engagement must be delivered without presuming incompetence on the part of women. Research indicates that women are more likely to encounter disrespectful instructor interactions that carry an undertone of presumed incompetence. These experiences can have lasting consequences, including reduced motivation to persist in computing (Hunt et al., 2022).

Similarly, our findings on grit underscore the necessity for customized support systems, the development of nurturing learning atmospheres, and a nuanced strategy for fostering grit while improving the persistence and accomplishments of women in computing programs. Research shows that hope, positive affect, and familial bonds can promote grit development (Christopoulou et al., 2018). Specifically, these factors contribute to a mindset that values effort, resilience, and perseverance, ultimately increasing an individual's likelihood of exhibiting grit in the face of challenges. Emphasizing these factors in how the introductory computer science sequence is taught can thus provide students with the mental and emotional resources necessary for long-term goal pursuit.

In addition, instructors should design assignments and tasks that better accommodate students who appear to have lower levels of grit. In practice, this is best operationalized as designing for productive persistence[4] rather than attempting to "fix" students' dispositions. Doing so shifts attention to instructor-controlled design choices that can sustain engagement, particularly for students who are more likely to disengage after setbacks. Specifically, instructors can increase course structure and incorporate active-learning routines (Eddy & Hogan, 2014; Freeman et al., 2014), make self-regulation explicit through guidance in planning, monitoring, and adjusting learning strategies (Donker et al., 2014; Theobald, 2021), and normalize setbacks as part of skill development through feedback and reframing that reduces withdrawal following difficulty (Dryden et al., 2021; Yeager et al., 2019). In addition, instructors can adopt

mastery-oriented assignment policies by using assessment data to tailor instruction and feedback, making progress visible and providing consistent, constructive guidance that supports both remediation and enrichment (Archambault et al., 2022). These measures offer an opportunity to reduce the impact of the grit deficit, ultimately supporting these students more effectively in their learning journey. At the same time, CS programs should make their learning environments more fully inclusive across gender, reducing reliance on "grit" as a gatekeeping trait and ensuring conditions are set for everyone to succeed. Together, these changes shift the burden from individual persistence to supportive pedagogy and climate.

Although the foregoing recommendations on programming experience and grit have primarily emphasized women, we recommend extending them to other underrepresented and marginalized students, including non-binary students and Women of Color. This broader framing is important because interventions designed as "gender-specific" solutions can unintentionally marginalize transgender and non-binary students when they implicitly essentialize a men – women binary that these students may not identify with (Menier et al., 2021). Accordingly, interventions should be designed and communicated in ways that support persistence across diverse gender and racial identities, as well as intersecting marginalized experiences.

7.3. Limitations and future work

As is common to self-reported surveys, our study might have suffered from non-response bias: students who chose not to participate in our survey may differ from those who did in ways that might impact the sample. On the other hand, the use of extra credit as an incentive may introduce response bias. We note that our survey experienced a remarkably high rate of response, which was likely influenced by the researchers' emphasis on the study's goal of improving the CS curriculum rather than the bonus credit.

Another limitation is the lower number of non-men participants. We note that the gender composition of our sample population is representative of the student population in the three courses we studied. This resulted in a lower but still useful proportion of women students and a very small number of students who identified as non-binary. This, unfortunately, required us to consider only the men and women students in our analyses. We acknowledge this as an inherent limitation of our survey research methodology and will aim to supplement our findings with qualitative research across all genders in future work.

Our study relies on cross-sectional survey data. Although our hypothesized directions are motivated by Social Cognitive Career Theory (SCCT), the design does not permit strong causal inference, and alternative (e.g. reciprocal) directions cannot be ruled out. Future work should employ longitudinal designs to test temporal precedence and reciprocal effects on the subjective experience. Experimental or quasi-experimental studies (e.g. efficacy-building instructional interventions) could further strengthen causal claims.

Relatedly, grit was assessed at a single time point using self-report items capturing diligence, effort regulation, and responses to difficulty. Although prior work indicates grit can be malleable (Hwang & Nam, 2021; Ma et al., 2020), our design does not permit inference about within-person change over time or the directionality between grit, programming experience, and SCCT beliefs. Longitudinal designs with repeated grit

measurement are needed to examine growth or change (and potential reciprocal relations) across course levels.

Our programming experience measure is course-referenced and self-reported. It captures the extent to which students perceive that their experience exceeds or falls short of the course's programming demands, rather than objective programming experience. Consequently, group differences may reflect differences in perceived preparedness or self-evaluation standards, and should not be interpreted as direct evidence of underestimation or overestimation of ability. Future work should triangulate this measure with objective indicators such as prior coursework, diagnostic assessments, and performance-based tasks.

Beyond measurement and design limitations, several substantive questions remain. Specifically, while our findings elucidate gender gaps across three course levels and identify their antecedents and consequences, we do not provide decisive evidence regarding the causes of the fluctuations in the identified gender gaps. Future research could take a longitudinal approach to unpack these fluctuations further, which could help clarify why gender gaps in grit narrow by CS3 and why women consistently report lower perceived programming experience across course levels despite overall gains at CS3. Such work can also examine why women's self-efficacy plateaus and declines at higher course levels while men's continues to rise. Ultimately, such work could answer important questions regarding the educational practices that can repair or prevent the observed gaps. In this light, one question is which course experiences, e.g. evaluation practices, workload, and classroom climate, best explain the CS1–CS3 reversal in women's grit and the widening confidence gap at CS3. A second question is whether perceived programming experience gaps primarily reflect differences in preparation or systematic differences in self-assessment standards and confidence estimation. Another question is whether targeted supports that build self-efficacy can prevent the CS3 downturn by improving confidence and downstream motivation, and whether the effects of such supports vary by course level and baseline experience.

This study is part of an ongoing long-term initiative aimed at discovering inclusive approaches to enhance achievement in computer science programs and explore methods to increase the retention, graduation rates, and post-graduation accomplishments of underrepresented student groups. It involves a partnership between two HBCUs and a PWI, representing a range of classifications within the Carnegie Classification of Institutions of Higher Education. At the time the data for the current study was collected, we had not yet achieved success in attaining a large sample of responses from the two participating HBCUs – future work based on the data from subsequent semesters can generalize the results from this paper across the three different institutions.

In addition to survey data, we have initiated data collection on students' performance and experience in programming at each course level across the three institutions. In future work, such data affords us opportunities to compare students' actual (rather than self-reported) performance and experience and to explore gender, racial, and institutional differences in performance. Moreover, we are conducting interviews with students and instructors at the three institutions to gain additional qualitative insights from both parties, which we will use to unpack further some of the findings of this paper in future work.

Ultimately, we aim for our research to inform inclusive pedagogical practices and community-based interventions that promote the retention of all students and provide tools for the computing education research community to design effective pedagogical interventions for underrepresented minority students in the major, leading to post-graduation success. The strengths identified through partnering institutions will be shared and promoted within and between institutions, and disseminated to other institutions. The work in the current paper is an initial step in this direction and has made important contributions in terms of our perpetual self-evaluation regarding our research approach.

8. Conclusion

This study describes the results of a survey of students in CS1, CS2, and CS3 at a PWI in the southeastern United States, studying the effect of differences in grit and programming experience on students' self-efficacy, outcome expectation, attitude toward the computing discipline, and intention to complete their CS degree. We find that despite the higher level of grit among women in CS1, and partially due to women's self-reported programming experience deficit across all course levels, women lag behind men in terms of self-efficacy, outcome expectation, attitude, and degree completion intention – all of which show stronger increases across course levels for men than for women. The identification of grit and experience as crucial factors determining the experiences of women in CS allowed us to formulate a number of strategies to improve their experiences in the future.

In particular, programs can reduce experience-driven disparities by expanding students' access to meaningful programming opportunities early, both before college and within gateway courses, rather than relying on self-selection into extracurricular experiences. Within CS1–CS3, instructors can intentionally design for sustained engagement by using relevant, choice-supported programming tasks, cultivating a supportive and stereotype-resistant classroom community, and strengthening confidence through frequent, improvement-oriented feedback and visible progress markers. Additionally, rather than treating grit as an individual "deficit", instruction can be organized to promote productive persistence through structured active learning, explicit support for self-regulation, and norms that frame setbacks as expected in learning to program. Finally, these strategies should be implemented in ways that support students across diverse gender identities and intersecting marginalized experiences, avoiding approaches that inadvertently reinforce a gender binary, and reducing reliance on grit as a prerequisite for success.

Notes

1. https://onlinedegrees.sandiego.edu/how-to-become-a-computer-scientist-salary-career-guide/.
2. Note that due to administrative limitations, the sample of survey data analyzed in this paper is limited to students at the PWI. We plan to revisit this work using data from all three institutions in the near future.
3. A non-significant χ^2 value indicates that the model is not significantly different from the observed data, implying a good fit. This is often desirable as it indicates that the hypothesized model adequately represents the data (Bentler & Bonett, 1980). The thresholds for alternative fit statistics are: CFI and TLI $\geq$ 0.95 and an RMSEA value of 0.06 or lower (Hu & Bentler, 1999).

4. *Productive persistence* is a concept introduced by the Carnegie Foundation for the Advancement of Teaching. It focuses on supporting students' beliefs about themselves as mathematics learners and practitioners, as well as their sense of belonging in mathematical settings.

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ORCID

Oluwafemi Osho http://orcid.org/0000-0001-8406-9467
Ananya Gupta http://orcid.org/0009-0000-3491-6453
Devon Brown http://orcid.org/0009-0006-3965-1566
Bart Kninjenburg http://orcid.org/0000-0003-1341-0669
Cazembe Kennedy http://orcid.org/0000-0002-7056-3111
Eileen Kraemer http://orcid.org/0000-0001-7331-1801
Stacey Sexton http://orcid.org/0000-0001-7287-1292
Gloria Washington http://orcid.org/0000-0002-8524-3559
John Porter III http://orcid.org/0000-0003-2534-3619
Kinnis Gosha http://orcid.org/0000-0001-8605-4672
Simone A. Smarr http://orcid.org/0009-0003-7600-0613

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Table A1. List of constructs and corresponding items.

Grit (alpha: 0.806)
- I am a hard worker.
- I am diligent.
- I can overcome obstacles to complete my tasks.
- I am a procrastinator.
- I give up when problems seem too difficult.

Programming Experience (alpha: 0.705)
- The coursework in X required less programming experience than I had.
- The coursework in X required more programming experience than I had.

Self-efficacy (alpha: 0.910)
- I am confident I will do well on computer science labs and projects.
- Even when I work hard at it, programming tends to be unusually hard for me.
- I am sure I can understand computer science.
- I am confident in my problem-solving ability.
- I am confident in my ability to meet unexpected programming challenges with success.
- I am confident in my ability to complete a major in computer science.
- I think I could handle difficult computing problems.
- I'm not good at computing.

Outcome expectation (alpha: 0.860)
- I expect that learning to use computer science skills will help me achieve my career goals.
- Programming is of no relevance to my life.
- I'll need a firm understanding of programming for my future work.
- Knowing programming will help me earn a living.
- I study programming because I know how useful it is.
- I see computer science as a subject I will rarely use in my daily life.
- My goals do not require that I learn computer science skills.
- I hope that in the future, I can find a career that does not require the use of computer science.

Attitude toward CS (alpha: 0.713)
- I would voluntarily participate in additional computer science activities outside of classes, if provided the opportunity.
- It would make me happy to be recognized as an excellent student in computer science.
- Being regarded as smart in computer science would be a great thing.
- Computer science is a worthwhile and necessary subject.
- I do not enjoy spending a lot of time writing programs.

Behavioral intention (alpha: 0.726)
- Complete a CS major or minor.
- Pursue a career after graduation where you directly apply the expertise and skills you will have acquired in your computing major or minor.
- Acquire professional certifications related to your computing major or minor.
- Consider a career where the knowledge and skills you will have acquired in your computing major or minor are not required.
- Change to a different major or minor.
- Pursue a graduate degree in computing.

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Appendix. Measurement scales

All items were measured on a 5-point Likert scale. The items in grey were excluded to obtain a maximum CA value.